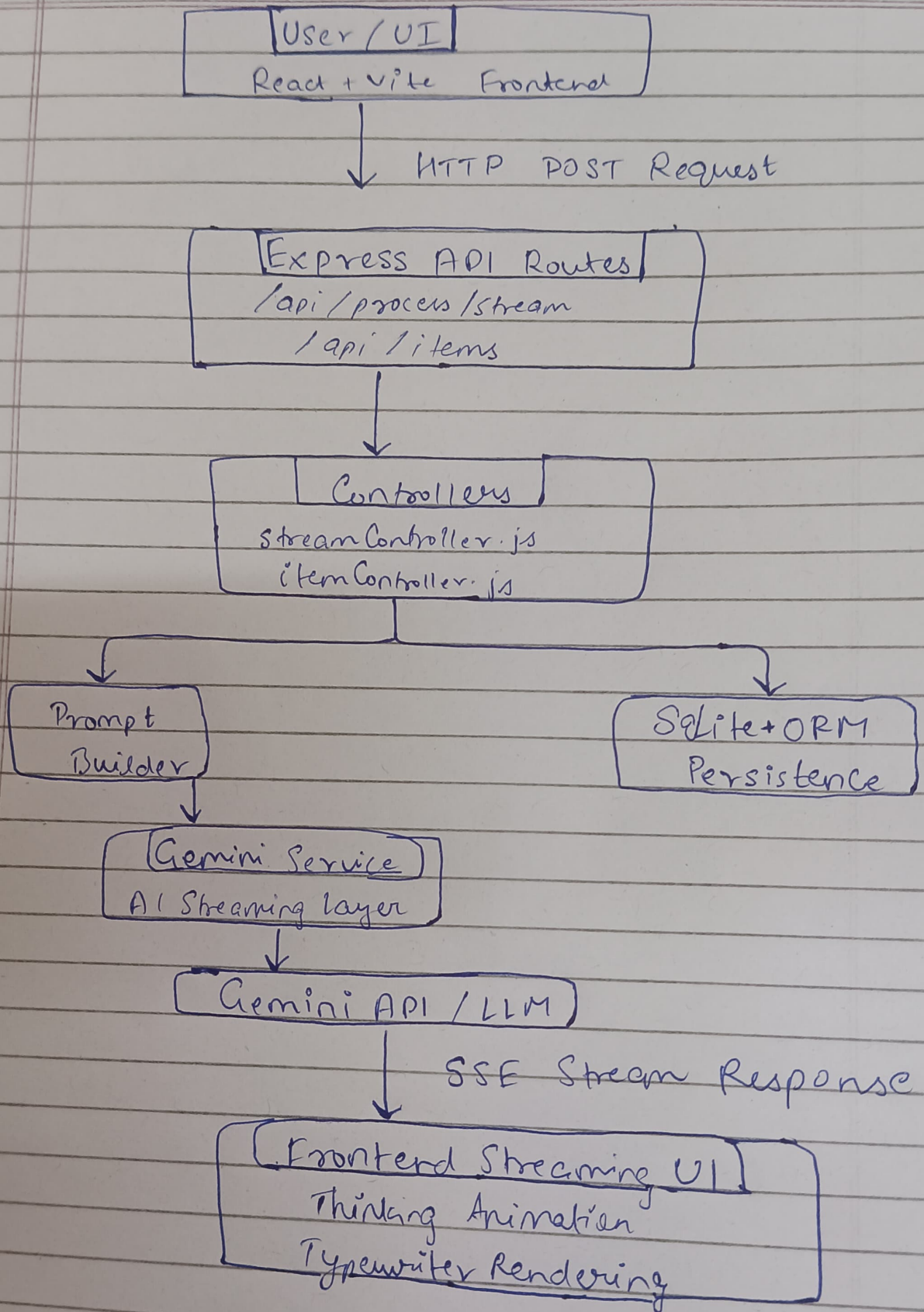


Architecture Diagram

Date: / /

Page No.



Complete Flow

Date: / /

Page No.

user enters text

• React frontend stores form state

user clicks Rewrite

Frontend sends POST request

↳ HTTPS post request to backend
API endpoint

↳ URL on backend that performs a specific task
eg - api/process/stream
task is to rewrite text using AI streaming

Express backend route receives request.

Controller catches request

prompt builder
Creates prompt,
gemini api called.

backend streams
LLM generated
chunks via SSE

Frontend receives chunks live

User sees streaming / typewriter effect

User saves to sidebar

Save rewrites appear in history.

Catches forwards it to the Controller

Stream Controller.js
receives rewrite request
• validates input
builds prompt
via rewrite prompt
starts Gemini streaming

Product - Full stack web application that uses LLM to do something useful with text user provides.

App Directions

1) Smart notes X (simple, summary)
can ↓ just AI.

2) Document Q&A - User uploads text.
- App lets them ask questions about content & returns ans grounded in the document

3) Content rewriter

useful for gmail

User pastes text, chooses tone/style.
App returns rewritten version & lets user save it.

↓
user can choose tone

↓ P & C

different groups.

Style & formality

Formal

Semi-formal

Conversational

Informal

Tone & emotion

• Professional

• Friendly ?

• Serious ?

• Emotional / Deep

• Persuasive /

Confident

Length / Detail

• One line / v. short

• Concise

• medium-length

• In-depth

user ~~can~~ can select 1 from each.

Then save.

Input Text



Choose : Formality

- Tone

- Length



Click rewrite

User gets feedback while
AI works

User saves result



Save history view (See like)

* Tech Stack

Frontend - • React + Vite

• Tailwind CSS

Backend - Node.js + Express

backend

on server

backend framework

running on

node.js

js runtime environment

Database - SQLite

ORM (Object-relational mapping)

↳ acts like bridge b/w

OOP languages &

relational DB's

Prisma

LLM - Gemini

↓
processes
↓
responds

Date: / /
Page: 10
happens
over HTTP.

IMP

* Prompt Module - prompts must be in dedicated modules.

* LLM Integration

- should stream responses

↓ ↳ token by token generation

use SSE - Server sent events

↓
One way communication of updates
to web pages from server to clients

→ simple UI - input box

- category boxes
- generated output
- save button
- ~~web~~ archive tab

* Edge cases - a) empty string

b) 1 emoji

c) 50k characters (keep limit 10k)

d) Hindi Input

e) Prompt Injection

↳ How to prevent?
↓

System prompt must
dominate